Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/03/2007
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Project Title:	Inflammation in the brain after particulate irradiation predisposes the hippocampus to a heightened vulnerability after a secondary insult		
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:	Add: Susanna Rosi, Ph.D. Co-I. Add: Marta Andres-Mach, Ph.D. Post-doctoral Fellow		
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Task Description:

A unique feature of the space radiation environment is the presence of high-energy charged particles including protons and fully ionized atomic nuclei such as 56Fe. These particulate radiations may pose a significant hazard to space flight crews not only during the mission but also later, on Earth, when slow-developing adverse effects finally become apparent. In the CNS, exposure to such radiation may not only directly affect structure/function within the brain, but it may also make tissues more sensitive to a secondary insult such as trauma, stoke or degenerative disease. We have shown that photon and particulate irradiations induce a strong and persistent inflammatory response in the brain that is associated with altered hippocampal neurogenesis and impaired cognitive function. Based on these data we hypothesize that such inflammatory changes, while they may only lead to minor morphologic/functional changes by themselves, will predispose the brain, and in particular hippocampal neurogenesis, to a heightened vulnerability after a secondary insult. Secondly, we hypothesize that a pharmacologic reduction of radiation-induced inflammation will ameliorate the combined effects of radiation and secondary injury on hippocampal neurogenesis. While the nature of possible secondary insults is variable, we have identified traumatic brain injury (TBI) as the choice for the proposed studies. We have identified 4 specific aims to test our hypotheses: 1) determine if inflammatory changes induced by a low dose of proton irradiation predisposes the hippocampus to abnormal neurogenesis after secondary TBI; 2) determine if inflammatory changes induced by a low dose of 56Fe irradiation predisposes the hippocampus to abnormal neurogenesis after secondary TBI; 3) using knock-out mice lacking chemokine receptor 2 (CCR2), determine if a muted monocytic inflammatory response can reverse the influence of low dose proton and/or 56Fe irradiation on TBI-induced alterations in neurogenesis; and 4) determine if pharmacologic intervention after proton and 56Fe irradiation will reduce radiation-induced inflammation and lessen the effects of subsequent TBI. Mice will be irradiated with low radiation doses (0–2 Gy) and 30 days later will subjected to mild/moderate TBI. Neurogenesis and inflammation will be detected using cell specific immunohistochemistry and the effects of irradiation will be quantified using confocal microscopy and unbiased stereology. These studies will address a possible mechanism by which low-dose irradiation may sensitize the brain to secondary injury, and they will also involve a pharmacologic approach to minimize such effects.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

To date, we have carried out 2 runs at BNL where we have given single doses of 56Fe irradiation (0, 0.5, 1, and 2 Gy) followed 1 month later by focal TBI. One month following TBI mice received a single injection of BrdU for 7 consecutive days; BrdU labels actively proliferating cells. Three weeks after the last BrdU injection mice were perfused and brain tissues were obtained for analyses of neurogenesis. The first run was in June, 2006, and each treatment group consisted of 8 mice. Tissues were collected in September of 2006 and of the original cohort of 32 animals we lost 5 after TBI. Tissues from the remaining mice were processed, sectioned and stained for either BrdU alone or from BrdU plus NeuN (neurons) and GFAP (astrocytes). Stereologic analyses are currently underway. The second cohort of mice was irradiated in October of 2006; tissues were collected in January of 2007. Those tissues have been processed, but sectioning is not yet underway. In the second group of 32 mice we lost 4 due to TBI. Because we had to wait for beam availability at BNL, we also initiated a study similar to that described above except that we used brain only x-rays and only two irradiation doses, 0 (controls) and 2 Gy, followed by TBI as described above. This study will provide important data against which the high LET iron data can be compared.
X-ray ± TBI Two-month old male C57BL/6 mice were used; 20 mice were randomly divided into groups of 5 mice each: sham treatment, TBI only, 0 Gy + TBI, and 2 Gy + TBI. Sham treated animals were anesthetized but not irradiated and had surgery (below) but no TBI. No complications were observed after irradiation and no mice were lost after TBI. During TBI, mice were anesthetized and placed in a stereotaxic frame . A circular craniotomy, 3.5 mm in diameter, was made in the left parietal skull, the dura was incised and mice were subjected to a controlled cortical impact. Briefly, a 3 mm diameter convex impact tip was used, and the impact was set to a 1.0 mm deformation delivered at a velocity of 4.5 m/sec. Sham-treated animals (controls) were subjected to the same surgical procedure but were not injured. Sixty days after irradiation (30 days after TBI) mice (n = 5) received a single i.p. injection (50 mg/kg) of BrdU daily for 7 days. Three weeks after the last BrdU injection, mice were anesthetized and perfused with ice-cold saline followed by freshly prepared, ice-cold 4% paraformaldehyde. The brain was removed, processed, and sectioned using a sliding microtome. To calculate the numbers of BrdU-positive cells in the dentate gyrus, at least 12 sections of a one-in-six series were scored per animal. All counts were limited to the dentate granule cell layer and a 50µm border along the hilar margin that included the dentate SGZ. Total numbers were obtained by multiplying the measured value by 6; overestimation was corrected for using a correction factor. Total numbers of BrdU-positive cells displaying the various lineage-specific phenotypes (neuron, astrocyte, activated microglia) were determined using confocal microscopy to score the co-localization of BrdU and phenotypic markers in representative sections from each animal.

RESULTS To date a total of 2-3 mice/treatment group have been analyzed. The results are shown below.

Total newly born cells: To assess the survival of newly born (BrdU+) cells, every sixth section through the entire dentate gyrus was stained for BrdU and all positively labeled cells counted in each hemisphere using a stereologic approach. After radiation only, there were decreases in the numbers of newly born cells in the ipsilateral and contralateral hemispheres relative to sham irradiated controls, and after TBI there were even greater decreases. After combined radiation and TBI the numbers of newly born cells were comparable to the values seen after TBI only and those values were lower than after radiation alone.

Newly born neurons; In sham treated controls the numbers of newly born neurons averaged about 1000 ± 100. Two Gy of x-rays reduced that number by about 20% in both hemispheres, while after TBI there was a much more substantial effect (i.e. ~ 60%). After combined treatment there were more new neurons produced relative to what was seen after TBI alone, suggesting that irradiation may have caused a preconditioning type of effect, whereby the effects of a subsequent traumatic injury were reduced relative to radiation alone.

Newly born astrocytes; The numbers of newly born astrocytes averaged about 25 in the ipsilateral hemisphere of sham treated animals. Radiation alone induced a 4 fold increase in astrocytes, probably due to a reactive gliosis. TBI resulted in only a minor increase. Combined treatment resulted in a very substantial increase (~8 fold) in newly born astrocytes. In the contralateral hemisphere, results were similar for x-ray and TBI alone, but the large increase after combined treatment was not evident.

	Microglia; Previous work by us has shown that decreased in neurogenesis resulting from x-irradiation or TBI is generally associated with increases numbers of newly born (BrdU+) activated microglia. In our preliminary study of combined injury we saw modest increases in total numbers of activated microglia in both hemispheres after irradiation but not after TBI. When the treatments were combined there were fewer microglia in the ipsilateral hemisphere, suggesting that the 2 treatments did not interact in terms of this endpoint . In the contralateral hemisphere the numbers of microglia after combined treatment were comparable to that seen after radiation alone. DISCUSSION The overall goal of this research proposal is to determine if low doses of particulate irradiation induce a monocytic inflammatory response in the brain and if that inflammation predisposes the hippocampal dentate gyrus to a heightened vulnerability upon a secondary injury, i.e. TBI. While the studies of heavy ion irradiation are still underway, we have been able to obtain some data regarding combined injury after low LET irradiation. While these data are preliminary they suggest that radiation impacts inflammatory processes and that there is reduced neurogenesis. The effects of TBI on neurogenesis are even more apparent. Surprisingly, when the 2 insults were combined, a month apart, there was no apparent additive effect. In fact, in terms of total new cells and newly born neurons, the irradiation appeared to precondition or 'protect' these cells upon subsequent TBI. At this time, the mechanism of this effect is not known. While we do not yet know if heavy ion irradiation will result in similar changes to what we have seen after x-rays, the latter data have highlighted a potentially important finding; that radiation may induce changes that may modulate the extent of injury induced upon a second insult. This could have important ramifications regarding potential strategies or approaches to the management of combined injury. Understanding the mechanism(s) responsible for the apparent 'preconditioning' effect could lead to potential pharmacologic interventions that could be used in the event of a post-irradiation traumatic event. Data from the next funding year should enable us to better understand this effect and determine if it is restricted to low LET irradiation or is more general in nature.
Bibliography Type:	Description: (Last Updated: 04/11/2018)
Articles in Peer-reviewed Journals	Rola R, Mizumatsu S, Otkuka S, Morhardt DR, Noble-Haeusslein, Fishman K, Potts MB, Fike JR. "Alterations in hippocampal neurogenesis following traumatic brain injury in mice." Exp Neurol. 2006 Nov;202(1):189-99. Epub 2006 Jul 28. PMID: 16876159 , Nov-2006
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